

MANY-ELECTRON ATOMS

AN EXERCISE WITH REALISTIC COMPLICATIONS

- We can solve the hydrogen-like problem (one particle in a known potential), but all real atoms and molecules are *many-body* systems.
- The true wave function for a N electron system is a $3N$ -dimensional (+ spin) object. It can only be treated for 2-3 particles.
- BUT if each electron can be described as moving in an *average* potential from all the other we are back to a one-particle problem!
- and further, if this *mean field* potential is centrally symmetric we just have a one-dimensional one-particle system to solve

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$$\left(-\frac{\hbar^2}{2m_e} \frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{2m_e r^2} - \frac{Ze^2}{4\pi\epsilon_0 r} + eV_{ee}(r) \right) P_{n\ell}(r) = E_{n\ell} P_{n\ell}(r)$$

- If you knew $V_{ee}(r)$ you could just plug it into your ready solution!

HOW TO FIND THE ELECTRONIC POTENTIAL

- Classical electromagnetic theory: The potential is given by the charge distribution:

$$\begin{aligned}\rho(r) &= e \sum_i^{N_{\text{occ}}} \int \frac{1}{4\pi} \Psi_i^*(r, \theta, \phi) \Psi_i(r, \theta, \phi) d\Omega \\ &= e \sum_j^{\text{occ. orb}} N_j \frac{1}{4\pi} \left(\frac{P_{n_j \ell_j}(r)}{r} \right)^2\end{aligned}$$

with N_j being the occupancy of orbital j . Note that

$$\int \rho(r) dV = 4\pi \int \rho(r) r^2 dr = N_{\text{occ}}$$

$$\nabla^2 V_{ee}^{\text{dir}}(r) = \frac{1}{r} \frac{\partial^2 \varphi(r)}{\partial r^2} = -\frac{4\pi \rho(r)}{4\pi \epsilon_0},$$

HOW TO FIND THE ELECTRONIC POTENTIAL

- Quantum mechanics: Electrons are indistinguishable and obey the Pauli principle

$$\Psi(1, 2) = -\Psi(2, 1)$$

The simplest wave function for e.g. two particles is written as a Slater determinant

$$\Psi(1, 2) = \frac{\phi_1\phi_2 - \phi_2\phi_1}{\sqrt{2}}$$

and we get the non-classical *exchange interaction*

$$\int \Psi(1, 2) H \Psi(1, 2) dV = \int (\phi_1\phi_2 H \phi_1\phi_2 - \phi_1\phi_2 H \phi_2\phi_1) dV$$

HOW TO FIND THE ELECTRONIC POTENTIAL

- Approximation based on the free electron gas: $V_{exch} \sim \rho^{1/3}$, Slater 1951 (full expression in the assignment sheet).
- Theoretical justification Kohn & Sham 1964: the ground state energy is a *functional* of the charge density.

SELF-CONSISTENCE

- Solve with $V_{ee} = 0$
- Calculate ρ
- Calculate V_{ee} (both direct and exchange) from ρ
- Solve with this V_{ee}
- Calculate a new ρ and a new V_{ee}
- repeat until the solution does not change any more, i.e. when the procedure is *self-consistent*.

WHAT TO INVESTIGATE

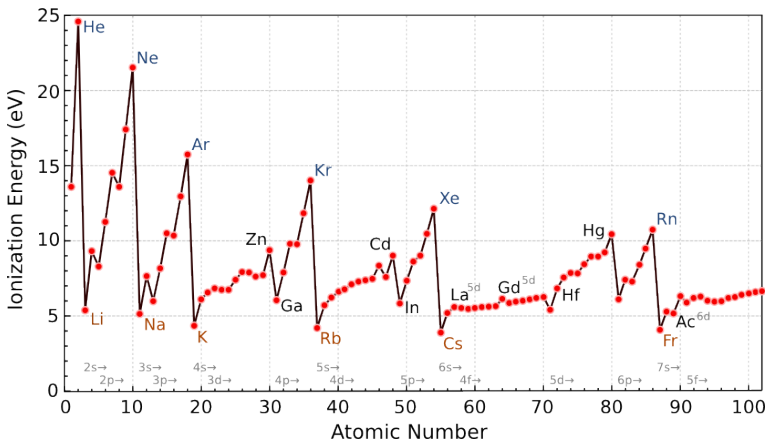


FIGURE: Your model should be able to explain the different ionization potential in the Periodic Table (picture from <https://commons.wikimedia.org/w/index.php?curid=93989781>)

WHAT TO INVESTIGATE

In which order are the shells filled? i.e. which *configuration* minimize the energy?

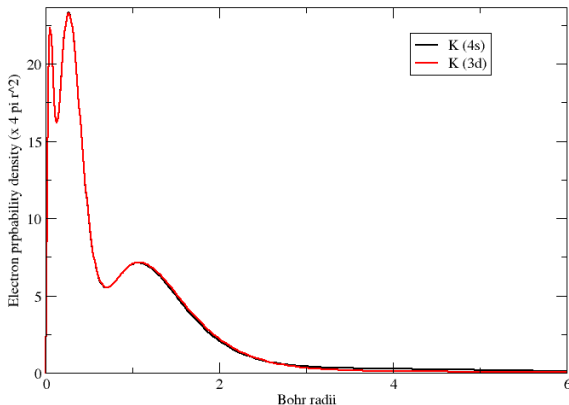


FIGURE: The plot shows $4\pi r^2 \rho(r)$, where $\rho(r)$ is the radial density for Potassium when the last electron is put in 4s or 3d.

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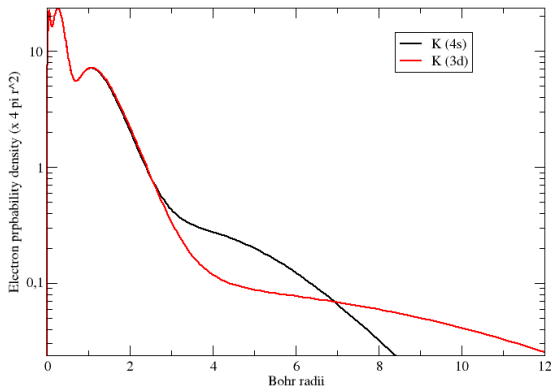


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SOME ADVICES

THE TOO HOT SHOWER PROBLEM

With iterative solutions it is common to *over compensate* and then the solution is oscillating between extremes! *Remedy*: Slow down the changes!

$$V_{ee}^{new}(r) \Rightarrow (1 - \eta)V_{ee}^{new}(r) + \eta V_{ee}^{old}(r), \quad (1)$$

where $0 < \eta < 1$, a typical value can perhaps be $\eta = 0.4$ Before you start the new iteration you redefine $V_{ee}^{old}(r)$

$$V_{ee}^{old}(r) \Rightarrow V_{ee}^{new}(r) \quad (2)$$

HOW TO CALCULATE THE IONIZATION ENERGY?

Each calculation gives an *orbital energy* for each quantum state $n\ell$. The orbital energy gives a first rough estimate of the binding energy.

- if all other orbitals stay *frozen* when one electron is ionized the orbital energy gives the ionization energy

But what you really want is the difference in total energy,

$$\Delta_{\text{ion}} = E_{\text{atom}} - E_{\text{ion}}.$$

- $E_{\text{total}} \neq \sum_i^{N_{\text{occ}}} E_{n\ell}^i$

Take care not to double count! The orbital energy for a includes the interaction with b and the orbital energy for b include the interaction with a

$$E_{\text{total}} = \sum_i^{N_{\text{occ}}} \left\{ E_{n\ell}^i - \frac{1}{2} \int P_{n\ell}^i(r) V_{ee}(r) P_{n\ell}^i(r) dr \right\}. \quad (3)$$

WHAT CAN GO WRONG?

- proper normalization needed when the electron density is to be evaluated from the orbital wave functions. Why? Check!
- The direct potential (from Poisson's eq.) at large distances $\sim +Q/r$. Why? What is Q ? Check!
- Total energy versus orbital energy somewhat tricky, be careful.
- Use the given results for helium (start with helium since it is easier) and neon to check your implementation. It is expected that the inner orbitals are only slightly changed by many-body effects (why?), but the outer are much more affected.

WHAT CAN GO WRONG?

- The inner orbitals get contracted in heavier elements. It is then important to have a knot sequence that can represent this contracted orbital.
- For example the peak of the hydrogen-like 1s probability density is found on a_0/Z . This affects of course also the potentials. See below:

